Homework 3 - solutions

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Problem 1 (Noether theorem & integrals of motion)

(**2 *points***) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric Coulomb potential

$$U(r,\, t) = \frac{U_0(t)}{r}, \quad U_0(t) = \frac{\kappa}{\sqrt{t}}, \quad \kappa > 0. \tag{1}$$

1. (**0.5 *point***) Identify all continuous symmetries of the action. Analyze if there are symmetries of type

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \;\; t \to \beta t, \qquad \alpha, \beta = \mathrm{const}$$

   which do not change the action or the equations of motion.

2. (**1 *point***) Construct the integrals of motion which might exist according to Noether's theorem. Demonstrate that these quantities are conserved using the equations of motion.

3. (**0.5 *point***) We have seen in lectures that in static (time-independent) Coulomb potential the particle's trajectory always remains in the plane (so-called scattering plane). Analyze if this property remains valid in the time-dependent potential given in Eq. (1). Analyze if the Kepler's laws remain valid in this potential field.

Problem 1 (Noether theorem & integrals of motion)

(**2 *points***) A pointlike nonrelativistic particle of mass m scatters in the time-dependent spherically symmetric Coulomb potential

$$U(r,\ t) = \frac{U_0(t)}{r}, \quad U_0(t) = \frac{\kappa}{\sqrt{t}}, \quad \kappa > 0. \tag{1}$$

1. The potential (lagrangian, action) are isotropically symmetric (invariant with respect to rotations around any axis)
   1. Noether theorem: conservation of all 3 components of angular momentum $\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{p}$
2. We can rewrite the potential as

   $$U(r,\ t) = \frac{\kappa}{r^2}\sqrt{\frac{r^2}{t}}, \tag{2}$$

   so there is a continuous symmetry (see next slide)

   $$\boldsymbol{r} \to \alpha \boldsymbol{r}, \qquad t \to \alpha^2 t, \qquad \alpha \in \mathbb{R}$$

3. There is no other continuous symmetries

Problem 1 (derivation of symmetry)

We can start with

$$\boldsymbol{r} \to \alpha \boldsymbol{r}, \qquad t \to \beta t, \qquad \alpha, \beta \in \mathbb{R}$$

The action transforms as

$$S \to S' = \int dt \left(\frac{\alpha^2}{\beta} \frac{m \dot{\boldsymbol{r}}^2}{2} - \frac{\sqrt{\beta}}{\alpha} \frac{\kappa}{r\sqrt{t}} \right)$$

- We should NOT forget extra "β" which comes from $\int dt$
- For invariance of equations of motion should request

$$\frac{\alpha^2}{\beta} = \frac{\sqrt{\beta}}{\alpha} \Rightarrow \beta = \alpha^2, \qquad \Rightarrow \frac{\alpha^2}{\beta} = \frac{\sqrt{\beta}}{\alpha} = 1$$

Note that for invariance of trajectory we start from requirement that constants in front of different terms coincide.

- We do not request that they equal to one: this is needed for Noether theorem, but not for invariance of equations of motion. However, in this problem we got that $S = \mathrm{const}$

Problem 1 (proof of invariance)

We analyzed symmetry

$$\boldsymbol{r} \to \alpha \boldsymbol{r}, \qquad t \to \alpha^2 t, \qquad \alpha \in \mathbb{R}$$

in Lecture 7, so we repeat almost literally those results. However, now energy $E \neq \text{const}$

$$S = \int dt \left(\frac{m\,\dot{\boldsymbol{r}}^2}{2} - \frac{\kappa}{r\sqrt{t}} \right)$$

Change of different quantities under rescaling:

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \;\; t \to \alpha^2 t, \qquad \alpha = e^{\lambda} \approx 1 + \lambda \qquad (1)$$

$$\frac{m\,\dot{\boldsymbol{r}}^2}{2} \to \frac{1}{\alpha^2} \frac{m\,\dot{\boldsymbol{r}}^2}{2}, \quad U(r) \to \frac{1}{\alpha^2} U(r)$$

$$\Rightarrow S = \int dt\, L \to \int \cancel{\alpha^2} dt\, \frac{1}{\cancel{\alpha^2}} L = \text{const}$$

This proves that (1) is indeed a continuous symmetry of the action.

Problem 1 (derivation of invariant)

To find the associated Noether invariant, consider $\alpha = \alpha(t) = e^{\lambda(t)}$, $|\lambda| \ll 1$ and pick up $\mathcal{O}(\dot{\lambda})$ term in the expansion of the action:

$$d\boldsymbol{r} \to (d\boldsymbol{r} + \boldsymbol{r} d\lambda) = \left(d\boldsymbol{r} + \boldsymbol{r}\,\dot{\lambda} dt\right), \qquad dt \to \left(1 + 2\dot{\lambda} t\right) dt$$

$$\boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{\boldsymbol{v} + \boldsymbol{r}\,\dot{\lambda}}{1 + 2\dot{\lambda} t} \approx \boldsymbol{v} + \dot{\lambda}\,(\boldsymbol{r} - 2t\,\boldsymbol{v})$$

$$S \equiv \int dt\, L \to \int dt\, L\,\left(1 + 2\dot{\lambda} t\right) + \int dt\, m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v})\,\dot{\lambda}$$

$$= \int dt\,[m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL]\,\dot{\lambda}$$

Problem 1 - Homework 3

$$Q = \frac{\partial \delta L}{\partial \dot{\lambda}} = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL =$$

$$= m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2t\left(\frac{m\,\boldsymbol{v}^2}{2} - U(r,t)\right)$$

$$= \boldsymbol{p} \cdot \boldsymbol{r} - 2\left(\frac{m\,\boldsymbol{v}^2}{2} + U(r,t)\right)\, t$$

$-$Check that $Q = \mathrm{const}$ using equations of motion and explicit expression for potential:

$$\frac{dQ}{dt} = \dot{\boldsymbol{p}} \cdot \boldsymbol{r} + \boldsymbol{p} \cdot \boldsymbol{v} - 2\left(\frac{m\,\boldsymbol{v}^2}{2} + U(r,t)\right) - 2t\left(\frac{\boldsymbol{p} \cdot \dot{\boldsymbol{p}}}{m} + \frac{\partial U(r,t)}{\partial t} + \boldsymbol{v} \cdot \frac{\partial U(r,t)}{\partial \boldsymbol{r}}\right)$$

$$= -\frac{\partial U}{\partial \boldsymbol{r}} \cdot \boldsymbol{r} + \cancel{\boldsymbol{p} \cdot \boldsymbol{v}} - 2\left(\cancel{\frac{m\,\boldsymbol{v}^2}{2}} + U(r,t)\right) -$$

$$- 2t\left(\cancel{-\frac{1}{m}\boldsymbol{p} \cdot \frac{\partial U(r,t)}{\partial \boldsymbol{r}}} + \frac{\partial U(r,t)}{\partial t} + \cancel{\boldsymbol{v} \cdot \frac{\partial U}{\partial \boldsymbol{r}} U(r,t)}\right) = \left[-\boldsymbol{r} \cdot \frac{\partial}{\partial \boldsymbol{r}} - 2 - 2t\frac{\partial}{\partial t}\right] U$$

$$= \left[-(-1) - 2 - 2 \cdot \left(-\frac{1}{2}\right)\right] U(r,t) = 0$$

as expected (!!!)

Problem 1 - Homework 3

1. Since $\boldsymbol{M} = \mathrm{const}$, repeating discussion from Lecture 7, conclude that trajectory always remains in plane $\perp$ to vector $\boldsymbol{M}$.
2. Since $|\boldsymbol{M}| = \mathrm{const}$, repeating discussion from Lecture 7, conclude that the 2nd law of Kepler remains valid
3. The first and the third laws of Kepler are NOT valid, since they use explicit trajectory for their proof.

Problem 2 - Homework 3

(***2 points***) A ***relativistic*** particle of mass m moves in the spherically symmetric potential $U(r) = kr^2/2$ of 3D harmonic oscillator.

−(***0.5 points***) Write out the lagrangian of the system, and the corresponding equations of motion

−(***0.5 points***) Identify all the symmetries of the action and construct all integrals of motion which might exist in this problem.

−(***0.5 points***) Try to integrate the equations of motion and find trajectories for bound orbits. Analyze if the trajectories are closed for bound orbits and if the motion remains periodic as a function of time.

−(***0.5 points***) Demonstrate that in the limit $c \to \infty$ your results reproduce well-known nonrelativistic limit. Assuming that c is large yet finite, find explicitly the first relativistic correction to trajectory.

Problem 2 - NR limit

—Start with nonrelativistic limit to develop qualitative understanding

$$L = \frac{mv^2}{2} - \frac{\kappa r^2}{2}$$

—analysis is staightforward in Cartesian coordinates since

$$\frac{\kappa r^2}{2} = \frac{\kappa\left(x^2 + y^2 + z^2\right)}{2}$$

—get 3 decoupled equations:

$$m\ddot{x} = -\kappa x,$$
$$m\ddot{y} = -\kappa y,$$
$$m\ddot{z} = -\kappa z,$$

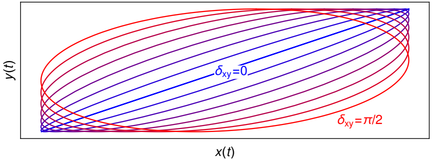
$$q_i = A_i \cos\left(\omega t + \varphi_i\right), \quad i = 1, 2, 3$$
$$q_i = \{x, y, z\}$$

—Potential is central ⇒motion is always in plane transverse to $\boldsymbol{M}$

—Assume $\boldsymbol{M} \sim \hat{\boldsymbol{z}}$, so motion occurs in xy plane

$$x = A_x \cos\left(\omega t + \varphi_x\right),$$
$$y = A_y \cos\left(\omega t + \varphi_y\right),$$
$$z = 0$$

—equation for ellpise $\delta_{xy} = \varphi_x - \varphi_y$ determines orientation:



*Different colors=different δ_{xy}

*$r = 0$ corresponds to center, NOT focus of the ellipse like in Kepler problem

Problem 2 - general relativistic case

−Lagrangian:

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} - U(r) = -mc^2\sqrt{1-\frac{\dot r^2 + r^2\left(\dot\theta^2 + \sin^2\theta\dot\phi^2\right)}{c^2}} - \frac{\kappa r^2}{2}$$

−generalized coordinates: r, θ, ϕ

−generalized momenta:

$$p_r = \frac{\partial L}{\partial \dot r} = \gamma m\,\dot r, \quad p_\theta = \frac{\partial L}{\partial \dot\theta} = \gamma m\,r^2\dot\theta, \quad p_\phi = \frac{\partial L}{\partial \dot\phi} = \gamma m\,r^2\sin^2\theta\dot\phi, \quad \gamma \equiv \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

−Explicit Euler-Lagrange equations:

$$\frac{d(\gamma m\,\dot r)}{dt} = -\kappa r + 2\gamma m r\left(\dot\theta^2 + \sin^2\theta\dot\phi^2\right),$$

$$\frac{d(\gamma m\,r^2\dot\theta)}{dt} = \frac{\gamma m r^2\sin 2\theta\dot\phi^2}{2}, \qquad \frac{d(\gamma m\,r^2\sin^2\theta\dot\phi)}{dt} = 0$$

−Last equation:$p_\phi = \mathrm{const}$, can use to eliminate $\dot\phi$:

$$\frac{d(\gamma m\,\dot r)}{dt} = -\kappa r + 2\gamma m r\left(\dot\theta^2 + \frac{p_\phi^2}{\gamma^2 m^2 r^4\sin^2\theta}\right), \quad \frac{d(\gamma m\,r^2\dot\theta)}{dt} = \frac{p_\phi^2\cos\theta}{\gamma m r^2\sin^3\theta}.$$

Problem 2 - Homework 3

–Lagrangian:

$$L = -mc^2\sqrt{1-\frac{v^2}{c^2}} - U(r) = -mc^2\sqrt{1-\frac{\dot{r}^2 + r^2\left(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2\right)}{c^2}} - \frac{\kappa r^2}{2}$$

–Explicit Euler-Lagrange equations are extremely complicated, so will use integrals of motion instead to find the trajectory

–**Symmetries** & integrals of motion:

$$\phi \to \phi + \phi_0 \Rightarrow p_\phi \equiv M_z = \mathrm{const}$$

$$t \to t + t_0 \Rightarrow E = \mathrm{const}$$

$$E = p_r \cdot \dot{r} + p_\theta \cdot \dot{\theta} + p_\phi \cdot \dot{\phi} - L =$$

$$= mc^2\gamma + \frac{\kappa r^2}{2}, \quad \kappa > 0$$

–In spherical coordinates we don't see other symmetries, but ...

Problem 2 - Homework 3

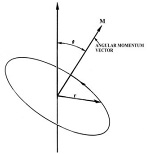
– In Cartesian coordinates we know that potential is invariant w.r.t. rotation around *any* axis

$$\Downarrow$$

All components of $\boldsymbol{M}$ are conserved, $\boldsymbol{M}$ keeps its direction

– In spherical coordinates the orientation of basis vectors $\boldsymbol{e}_\theta, \boldsymbol{e}_\phi$ changes, so we don't see this property (actually there is a hidden symmetry which mixes θ, ϕ in non-trivial way)

Consequences of $\vec{M} = \mathrm{const}$:



- *Particle <u>always</u> moves in a plane ortogonal to $\vec{M}$, i.e. vectors $(\vec{r}(t), \vec{v}(t))$ always remain in this plane, for $\forall t$.*

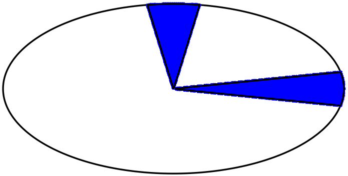
– We may choose plane $\theta = \pi/2$ as a plane of motion, like in nonrelativistic case

$$M \equiv \gamma m r^2 \dot{\phi} = \frac{m r^2 \dot{\phi}}{\sqrt{1 - v^2/c^2}} = \mathrm{const}$$

<u>Second law of Kepler is NOT valid</u>

– Due to $\gamma \neq 1$, see that $r^2 d\phi \neq \mathrm{const}$ for relativistic particle in general.

– It works ONLY for special case when particle has $v \ll c$ or $v = \mathrm{const}$ (energy conservation: this is possible only along circular orbit with $r = \mathrm{const}$)



Problem 2 - Homework 3

- Energy conservation (motion plane $\theta = \pi/2$ orthogonal to $\boldsymbol{M}$)

$$E = m\gamma c^2 + U(r)$$

$$\gamma = \left[1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right]^{-1/2}$$

$$M_z = m\gamma r^2\dot{\phi} = \mathrm{const}$$

$$\Rightarrow M_z^2\left(1 - \frac{\dot{r}^2 + r^2\dot{\phi}^2}{c^2}\right) = m^2 r^4 \dot{\phi}^2$$

$$\dot{\phi} = \frac{M_z}{mr^2}\sqrt{\frac{1 - \dot{r}^2/c^2}{1 + M_z^2/(m^2r^2c^2)}}$$

$$\gamma = \frac{M_z}{mr^2\dot{\phi}} = \sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}}$$

$$E = mc^2\sqrt{\frac{1 + (M_z/mrc)^2}{1 - \dot{r}^2/c^2}} + \frac{\kappa r^2}{2} \quad (1)$$

In nonrelativistic case ($c \ll 1$) this result yields

$$E_{\mathrm{nonrel}} = E - mc^2 = \quad (2)$$
$$m\dot{r}^2 + \frac{M_z^2}{mr^2} + \frac{\kappa r^2}{2} + \mathcal{O}\left(\frac{1}{c^2}\right)$$

and we interprete the last 2 terms as an effective potential.

– The particle never reaches $r = 0$ nor $r = \infty$ in the nonrelativistic limit, since potential U goes to infinity there.

Problem 2 - Homework 3

$$E = mc^2\sqrt{\frac{1+(M_z/mrc)^2}{1-\dot{r}^2/c^2}} + \frac{\kappa r^2}{2} \tag{1}$$

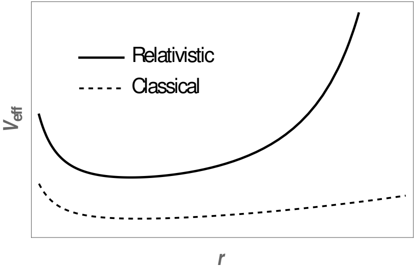
– We'll try to rewrite (1) as:

$$\frac{mc^2}{2} = \frac{m\dot{r}^2}{2} + \frac{m^3c^6}{2}\left(1+\left(\frac{M_z}{mrc}\right)^2\right) \Big/ \left(E-\frac{\kappa r^2}{2}\right)^2 \tag{2}$$

Can interpret the last term as "potential" and use it for qualitative interpretation, however there is some peculiarity:

* "Potential" depends on true energy E
* Should consider mc^2 as "energy" (see left-hand side)
* Large-c expansion near $E \approx mc^2$

$$V_{\text{eff}} \approx \frac{\kappa r^2}{2} + \frac{M_z^2}{2mr^2} + \frac{\kappa\left(4M_z^2 + 3m\kappa r^4\right)}{8m^2c^2}$$



Problem 2 - Homework 3

Solution of EOMs:

$$\frac{dr}{dt} = f(r, E) = \sqrt{c^2 - m^2 c^6 \left(E - \frac{\kappa r^2}{2}\right)^{-2} \left(1 + \left(\frac{M_z}{mrc}\right)^2\right)}$$

$$\int^r \frac{dr}{f(r, E)} = t - t_0 \quad (3)$$

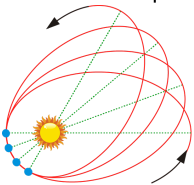
The integral over r is expressed in terms of elliptic functions.

$-$Equation for trajectory $r(\phi)$:

$$\frac{d\phi}{dr} = \frac{\dot{\phi}}{\dot{r}} = \frac{M_z}{mc^2\, f(r,\ E)} \left(E - \frac{\kappa r^2}{2}\right) \equiv g(r, E)$$

$$\Rightarrow \phi - \phi_0 = \int dr\, g(r, E)$$

*Trajectories are NOT ellipses, NOT even closed.



Corrections to NR trajectory

−jump back to Cartesian coordinates (easier analysis)
Effective potential:

$$\tilde{V}_{\text{eff}} \approx \frac{\kappa r^2}{2} + \frac{\kappa\left(4M_z^2 + 3m\kappa r^4\right)}{8m^2c^2} \approx \frac{\kappa\left(x^2+y^2+z^2\right)}{2} + \frac{\lambda}{4}\left(x^2+y^2+z^2\right)^2$$

−grey-colored term does not effect EOM
Equations of motion:

$$m\ddot{x} = -\kappa x + \lambda x\left(x^2+y^2+z^2\right),$$
$$m\ddot{y} = -\kappa y + \lambda y\left(x^2+y^2+z^2\right),$$
$$m\ddot{z} = -\kappa z + \lambda z\left(x^2+y^2+z^2\right),$$

Solution of EOMs (perturbative approach):

$$x \approx A_x \cos\left(\omega t + \varphi_x\right) + \lambda\, x_1(t) + \lambda^2 x_2(t) + \ldots$$
$$y \approx A_y \cos\left(\omega t + \varphi_y\right) + \lambda\, y_1(t) + \lambda^2 y_2(t) + \ldots$$
$$z \approx 0$$

$$\ddot{x}_1 + \omega_0^2 x_1 \approx \lambda\left(A_x^3 \cos^3\left(\omega t + \varphi_x\right) + A_x A_y^2 \cos\left(\omega t + \varphi_x\right)\cos^2\left(\omega t + \varphi_y\right)\right) = f_x(t)$$
$$\ddot{y}_1 + \omega_0^2 y_1 \approx \lambda\left(A_y A_x^2 \cos\left(\omega t + \varphi_y\right)\cos^2\left(\omega t + \varphi_x\right) + A_y^3 \cos^3\left(\omega t + \varphi_y\right)\right) = f_y(t)$$

Corrections to NR trajectory

$$\ddot{x}_1 + \omega_0^2 x_1 \approx \lambda \left(A_x^3 \cos^3(\omega t + \varphi_x) + A_x A_y^2 \cos(\omega t + \varphi_x) \cos^2(\omega t + \varphi_y) \right) = f_x(t)$$

$$\ddot{y}_1 + \omega_0^2 y_1 \approx \lambda \left(A_y A_x^2 \cos(\omega t + \varphi_y) \cos^2(\omega t + \varphi_x) + A_y^3 \cos^3(\omega t + \varphi_y) \right) = f_y(t)$$

Solution:

$$x_1(t) = \int^t d\tau \, G(t-\tau) \, f_x(\tau), \quad y_1(t) = \int^t d\tau \, G(t-\tau) \, f_y(\tau),$$

where G is Green function

$$G(\xi) = \int \frac{d\omega}{2\pi} \frac{e^{i\omega\xi}/m}{(-\omega^2 + \omega_0^2)} = \frac{\theta(\xi)}{m\omega_0} \sin(\omega_0 \xi)$$

Similarly can restore term-by-term higher order corrections